MILITARY SPECIFICATION

BOOSTER, FUZE, MARK 60 MOD 1

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the MARK 60 MOD 1 Fuze Booster critical item, referred to herein as the booster.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification for.
MIL-P-5425	Plastic, Sheet, Acrylic, Heat Resistant.
MIL-C-21723	Composition CH-6.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-C-45662	Calibration System Requirements.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-453	Inspection, Radiographic.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1695	Environments, Working, Minimum Standards for.

PUBLICATIONS

MANUALS

DoD 4145.26M	DoD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

DL 1137AS116	Booster, Fuze, MARK 60 MOD 1, Data List.

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DL 1137AS292	Lead, Explosive, MARK 25 MOD 0, Data List.
LD 533567	Detonator, Electric, MARK 87 MOD 0, List of Drawings.
SA 2747612	Fixture, Vibration, Three Plane.
SA 2747613	Fixture, Jolt Test.
SA 2747614	Fixture, 40-Foot Drop.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS.

3.1 Item definition. The booster initiates the warhead of a guided missile and is composed of a shell, cover, mounting flange, and an explosive pellet containing approximately 116 grams of Composition CH-6 explosive in accordance with MIL-C-21723.

3.1.1 Government-furnished property. When specified in the contract or purchase order, the Government will furnish the contractor with Composition CH-6 explosive in a quantity sufficient to meet contract requirements (see 6.2.1).

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Function. The booster shall function as required when initiated by a MARK 87 MOD 0 Electric Detonator (LD 533567) and a MARK 25 MOD 0 Explosive Lead (DL 1137AS292), respectively.

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3.2.1.2 Function and output. The depth of the dent shall be 0.200 inch minimum when measured in accordance with Figure 1.

3.2.2 Environmental conditions. The booster shall not be damaged, nor shall safety or subsequent performance be degraded, by exposure to any natural combination of service-use environments. These shall include, but shall not be limited to, the following environments.

3.2.2.1 Transportation vibration-temperature (nondestructive). The booster shall meet the requirements of MIL-STD-331, Test 119, Procedure 2. The booster shall function with the proper output, and show no evidence of cracks, dents, or other deformities after subjection to the environment of 4.4.3.1.

3.2.2.2 Temperature and humidity (nondestructive). The booster shall meet the requirements of MIL-STD-331, Test 105, Two Chamber Method. The booster shall function with the proper output, and show no evidence of cracks, ruptures, or other deformities after subjection to the environment of 4.4.3.2.

3.2.2.3 Random vibration (nondestructive). The booster shall meet the requirements of MIL-STD-810, Method 514.2, Procedure V, Part 2, Curve AH. The booster shall function with the proper output, and show no evidence of cracks, dents, or other deformities after subjection to the environment of 4.4.3.3.

3.2.2.4 Forty-foot drop (destructive). The booster shall meet the requirements of MIL-STD-331, Test 103, Procedure 2. The booster shall not explode nor burn during the test, and shall be safe to handle and dispose of after subjection to the environment of 4.4.3.4.

3.2.2.5 Jolt (destructive). The booster shall meet the requirements of MIL-STD-331, Test 101, Procedure II. The booster shall be safe to store, handle, and transport, and shall show no evidence of cracks, ruptures, or other deformities after subjection to the environment of 4.4.3.5.

3.2.3 Radiographic. The booster shall show no evidence of loose powder, voids, fissures, or cracks, and all parts shall be properly located.

3.3 Design and construction.

3.3.1 Production drawings. The booster shall meet the requirements of and be manufactured and assembled in accordance with the drawings, parts lists, and other documents listed on DL 1137AS116.

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3.3.2 Standards of manufacture.

3.3.2.1 Facilities and working environment. The contractor shall provide adequate facilities for the fabrication, assembly, and testing of the booster in accordance with MIL-STD-1695(AS).

3.3.2.2 Safety precautions. Unless otherwise specified in the contract or purchase order, the safety precaution requirements of DoD 4145.26M shall apply (see 6.2.1).

3.3.3 Test equipment calibration. Test equipment shall be calibrated and maintained in accordance with a plan meeting the requirements of MIL-C-45662. When specified in the contract or purchase order, the calibration plan shall be submitted to the Government for approval prior to use (see 6.2.2).

3.3.4 Test plan and test procedures. When specified in the contract or purchase order and using MIL-STD-831 as a guide, the contractor shall prepare and submit to the Government for approval, a test plan and test procedures prior to testing (see 6.2.2).

3.4 First article.

3.4.1 Preproduction sample. Unless otherwise specified in the contract or purchase order, the contractor shall furnish a preproduction sample of 30 boosters to the testing activity designated in the contract or purchase order for first article inspection and approval (see 4.1.2.1 and 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample is at the risk of the contractor.

3.5 Workmanship. Workmanship during manufacture of the booster shall meet the requirements of MIL-A-2550. Workmanship shall not be limited solely to MIL-A-2550 and shall not preclude the use of time- and labor-saving techniques commensurate with the expendable nature of the booster, provided that performance and safety are in no way jeopardized.

3.5.1 Final visual. The fully assembled booster shall show no external evidence of cracks, burrs, dents, loose powder, or other deformities which could interfere with or prevent proper functioning (see 3.2.1.1).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in

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the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) Preproduction inspections (see 4.1.2.1).

(b) Quality conformance inspections (see 4.2).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction inspections. A preproduction sample of 30 boosters shall be subjected to the tests and examinations of Figure 2. The failure of any sample booster to meet all the requirements of Section 3 shall cause rejection of the preproduction sample.

4.2 Quality conformance inspections. All boosters of the production lot shall be subjected to the tests and examinations of Figure 3.

4.2.1 Inspection lot. The term inspection lot, as applied to Government inspection of boosters, shall mean all boosters of a production lot submitted at one time for acceptance. All boosters of the inspection lot shall be alike in construction and from the same production facility.

4.2.2 Sampling. Sampling and sample size shall be in accordance with MIL-STD-414, Inspection Level IV, Standard Deviation Method, Variability Unknown, and an AQL of 0.65.

4.3 Test equipment and conditions.

4.3.1 Standard test equipment. The contractor shall provide and maintain an adequate system of inspection and test equipment necessary to ensure that parts and products will meet the contract or purchase order, specification, and drawing requirements. Test equipment, including associated calibration programs and operation and maintenance procedures, is subject to approval by the procuring activity. When special test equipment and circuits are devised or commercially

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available equipment is employed, all test equipment circuits and methods are subject to approval by the procuring activity. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the tolerance of the requirement being measured.

4.3.1.1 Test equipment calibration. Test equipment shall be calibrated and maintained as specified in 3.3.3.

4.3.1.2 Gages. The contractor shall provide whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. When specified in the contract or purchase order, the Government will furnish drawings of pertinent Government final inspection gages (see 6.2.1). However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

4.3.2 Test conditions. Unless otherwise specified herein, the test methods of 4.4 shall be performed under the conditions specified in 3.3.2.1.

4.3.3 Test plan and test procedures. The contractor shall prepare a test plan and test procedures as specified in 3.3.4.

4.3.4 Inspection record. The inspection record shall include the results of all tests and inspections specified herein and any additional tests and inspections of any part or end item deemed necessary by the contractor to achieve selection or acceptance. The record shall be maintained by the contractor, organized to facilitate search, and readily available for review.

4.4 Methods of examination and test.

4.4.1 Drawing inspections. Drawing inspections shall be performed on each lot of in-process parts or subassemblies and the end item booster in accordance with DL 1137AS116. To be acceptable, the booster shall meet the requirements of 3.3.1.

4.4.2 Function and output test.

WARNING

The function and output test involves hazardous operations and requires

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the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews (see 3.3.2.2).

The function and output test consists of detonating the booster on a steel dent block and measuring the depth of the dent to determine the output. To be acceptable, the booster shall meet the requirements of 3.2.1.2.

4.4.2.1 Test method. Using the test setup of Figure 4, with the lead holder and ring test assembly of Figure 5, the booster shall be subjected to the function and output test as follows:

WARNING

The booster causes a significant blast and fragment hazard. Verify that the area is cleared of all personnel before applying energy to the firing circuit (see 3.3.2.2).

(a) Remove shorting plug from the firing circuit and verify continuity of the circuit.

(b) Apply firing energy to the test setup assembly (Figure 4) using the power supply specified for the MARK 87 MOD 0 Electric Detonator.

4.4.2.2 Output dent measurement procedure.

CAUTION

Do not change the relative position of the dent block on the support blocks after zeroing the gage indicator.

After firing, remove all foreign materials from the dent block. With the dent block resting on two parallel support blocks as shown on Figure 1, place the gage probe of a dial-indicator height gage in the deepest part of the dent and zero

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the gage indicator. Remove the probe from the dent and take gage readings at two points, each 1/8 inch from the opposite edge of the dent block as shown on Figure 1. The average of the two readings shall be considered the depth of the dent. The lower limit of the booster output, which shall be used in the computation of MIL-STD-414, shall be 0.214 inch. The data obtained from this test shall be used to compute the quantity (X-L)/S, as defined by MIL-STD-414, Single Specification Limit, Form 1, Variability Unknown, Standard Deviation Method. The lot shall be accepted if the quantity (X-L)/S is equal to or greater than 2.00 inches. If the quantity is less than 2.00 inches, the lot shall be rejected. In addition, a single dent of less than 0.200 inch from any one booster shall cause rejection of the lot.

4.4.3 Environmental.

4.4.3.1 Transportation vibration-temperature (nondestructive). The booster shall be tested in accordance with MIL-STD-331, Test 119, Procedure 2, and as follows:

(a) Mount the vibration fixture of SA 2747612 on a vibration table, and tighten the hold-down bolts to a torque of 20 ±1 foot-pounds (ft-lb).

(b) Mount the booster to the vibration fixture, and tighten the mounting screws to a torque of 17 ±1 inch-pounds (in-lb).

To be acceptable, the booster shall meet the requirements of 3.2.2.1.

4.4.3.2 Temperature and humidity (nondestructive). The booster shall be tested in accordance with MIL-STD-331, Test 105, Two Chamber Method. To be acceptable, the booster shall meet the requirements of 3.2.2.2.

4.4.3.3 Random vibration (nondestructive). The booster shall be tested in accordance with MIL-STD-810, Method 514.2, Procedure V, Part 2, Curve AH, and as follows:

(a) Mount the vibration fixture of SA 2747612 on a vibration table, and tighten the hold-down bolts to a torque of 20 ±1 ft-lb.

(b) Mount the booster to the vibration fixture, and tighten the mounting screws to a torque of 17 ±1 in-lb.

(c) Vibrate the booster along each of the three mutually perpendicular axes (vertical, horizontal, and longitudinal).

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NOTE: To prevent damage to test equipment, limiting of instantaneous acceleration peaks to three times the overall root-mean-square acceleration level is permitted.

To be acceptable, the booster shall meet the requirements of 3.2.2.3.

4.4.3.4 Forty-foot drop (destructive). Mount the booster to the test fixture of SA 2747614 and tighten the hold-down screws to a torque of 17 ±1 in-lb. The booster shall be tested in accordance with MIL-STD-331, Test 103, Procedure 2. To be acceptable, the booster shall meet the requirements of 3.2.2.4.

4.4.3.5 Jolt (destructive). The booster shall be tested in accordance with MIL-STD-331, Test 101, Procedure II, and as follows:

(a) Mount the jolt fixture of SA 2747613 to the jolt machine, and tighten to a torque that will prevent loosening during the test.

(b) Mount the booster to the jolt fixture, and tighten the hold-down screws to a torque of 17 ±1 in-lb.

To be acceptable, the booster shall meet the requirements of 3.2.2.5.

4.4.4 Radiographic inspection. Using radiographic equipment capable of detecting assembly defects such as cracks, voids, fissures, or loose powder, perform a radiographic inspection of each booster in accordance with MIL-STD-453. A minimum of two X-rays shall be made of each booster, each 90 degrees apart and showing the longitudinal section through the booster. To be acceptable, the booster shall meet the requirements of 3.2.3.

4.4.5 Final visual inspection. Prior to packaging, each booster shall be visually inspected. To be acceptable, the booster shall meet the requirements 3.5.1.

4.5 Packaging, packing, and marking inspection. Prior to shipment, the packaging, packing, and marking of the booster shall be inspected to ensure conformance to the requirements of Section 5.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be Level C in accordance with FED-STD-102.

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5.1.1 Level C. Boosters shall be clean, dry, and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level C in accordance with FED-STD-102.

5.2.1 Level C. Boosters, packaged in accordance with 5.1.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Packing shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with MIL-STD-129 (see 6.2.1).

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The booster is intended for use in the AIM-54 Phoenix missile.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Number, title, and date of this specification.

(b) Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.1.1.

(c) Safety precautions, if other than as specified in 3.3.2.2.

(d) When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 30 boosters as specified in 3.4.1 and 4.1.2.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

(e) Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

(f) Government final inspection gage drawings (see 4.3.1.2).

(g) Special marking (see 5.3).

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6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited as follows:

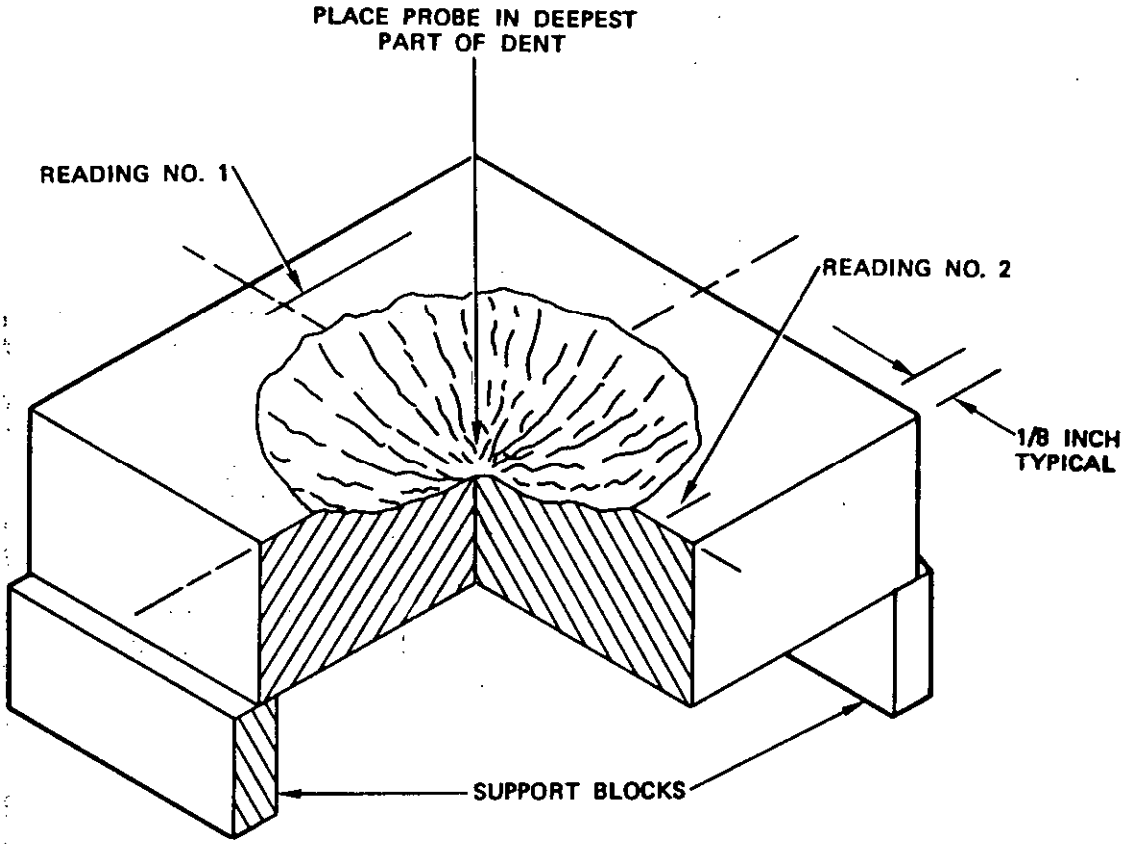
Paragraph	Data Requirement	Applicable DID
3.3.3	Calibration Plan	UDI-T-23742A
3.3.4	Test Plan and Test Procedures	DI-T-5204

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Preparing Activity:
Navy - AS

(Project 1336-N334)

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NOTES:

1. DENT BLOCK: STEEL, COLD FINISHED BAR IN ACCORDANCE WITH ASTM–A 108 GRADE 1018 OR 1020, ROCKWELL. HARDNESS B70 TO B95, SURFACE CONDITION AS ROLLED. SIZE, 5,000 ±0.125 INCHES SQUARE, 1.5 STOCK. BLOCK SHALL BE CLEAN AND FREE FROM RUST AND BURRS.
2. SUPPORT BLOCKS: WIDTH, 1/2 INCH MAXIMUM. LENGTH, APPROXIMATELY 5 INCHES. HEIGHT, OPTIONAL. ALLOY AND COMPOSITION, OPTIONAL. SUPPORT BLOCKS SHALL BE RIGIDLY CONNECTED SO THAT THEIR CENTERLINES ARE 5 INCHES APART.

FIGURE 1. Output dent measurement.

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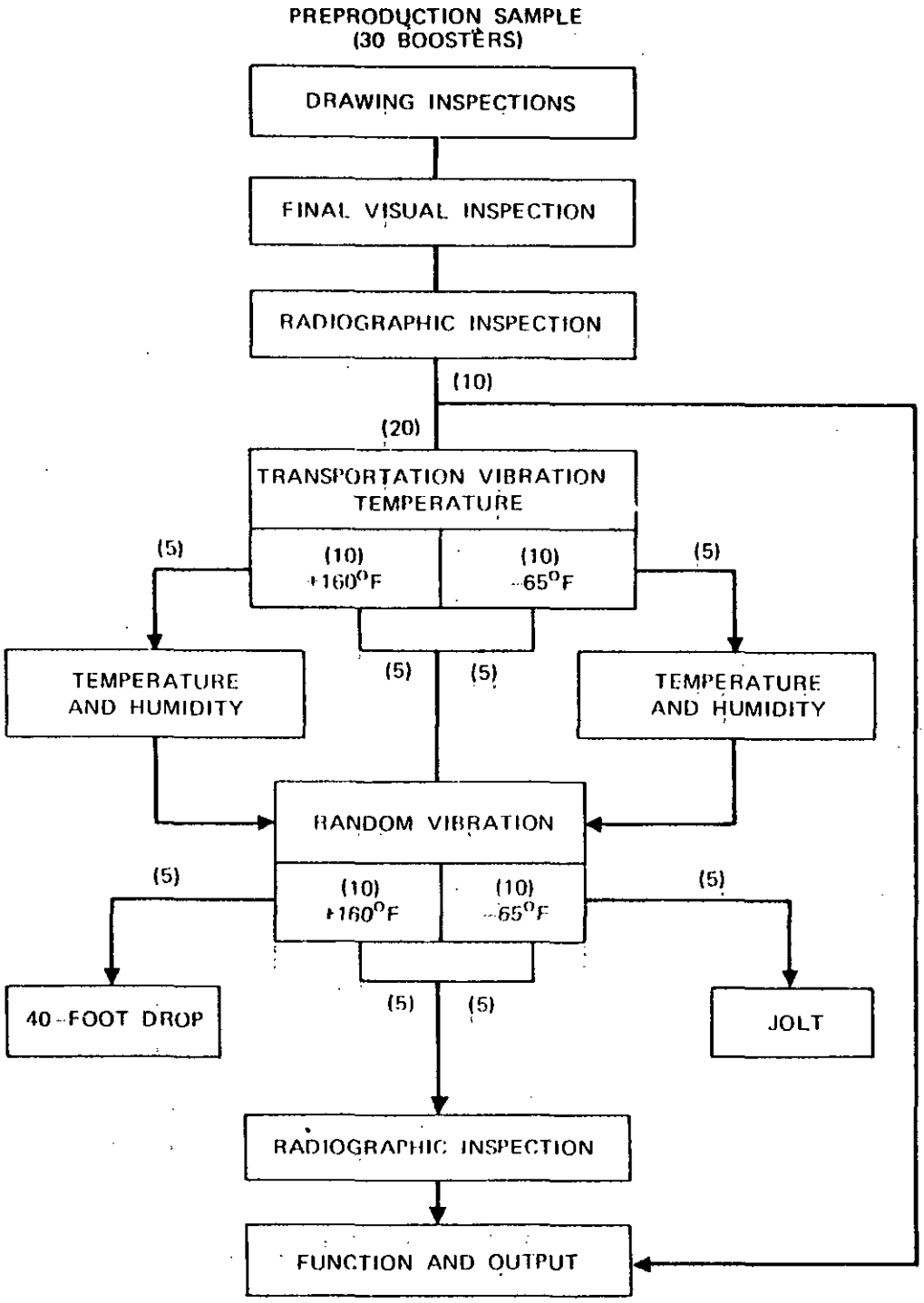


FIGURE 2. Preproduction sample test and examination sequence.

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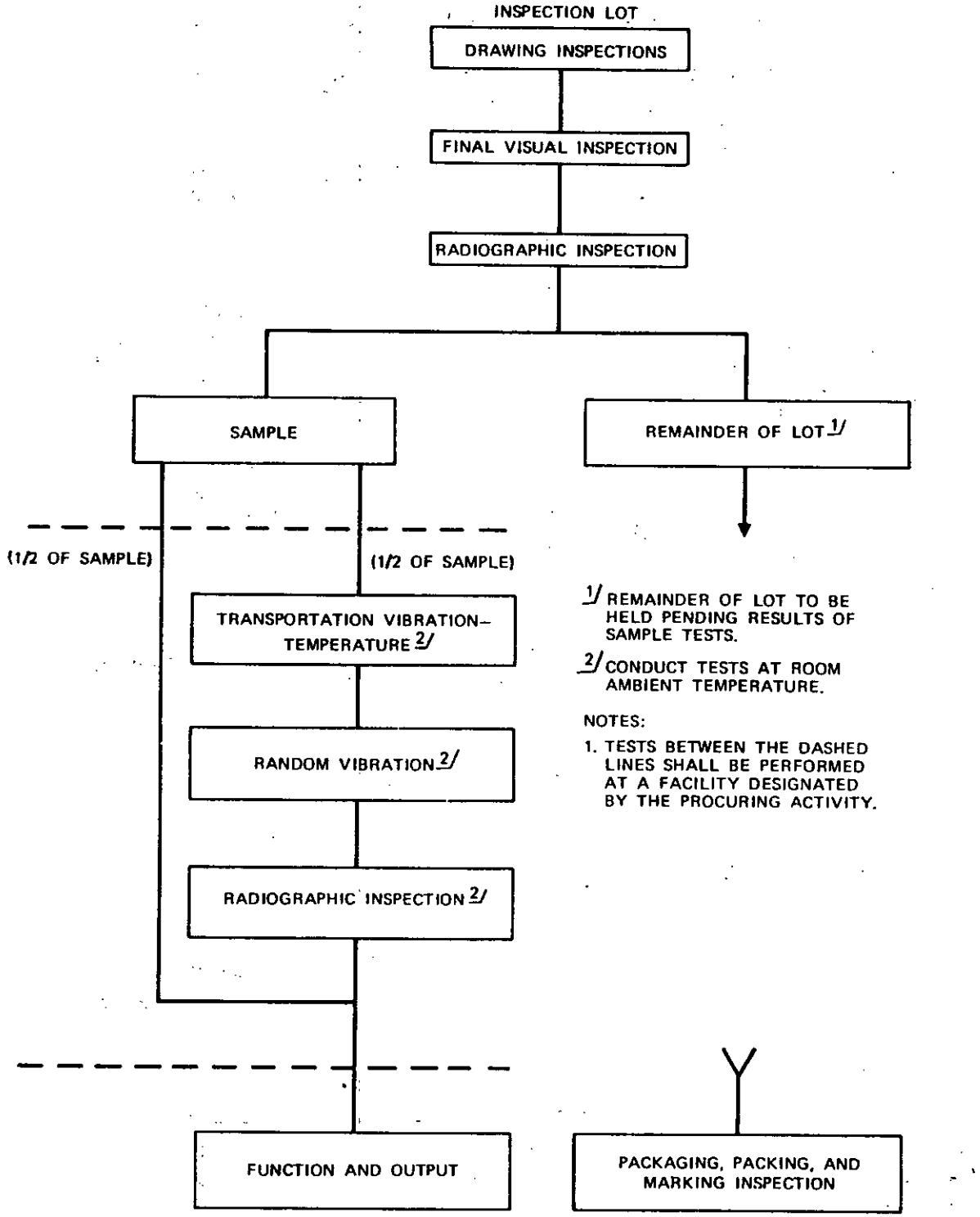
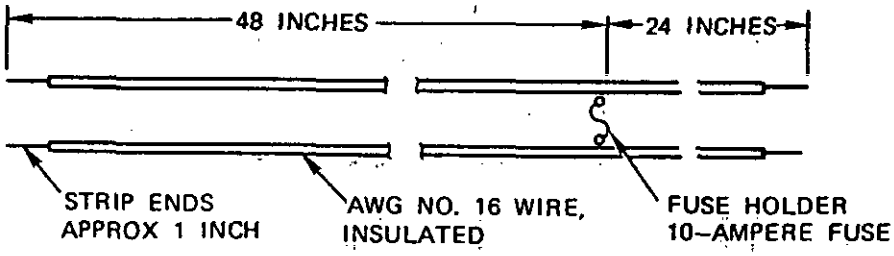
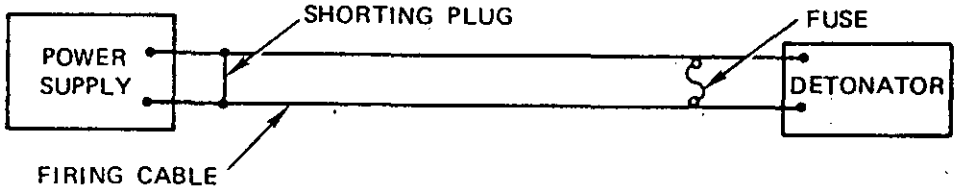


FIGURE 3. Quality conformance test and examination sequence.

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DETAILS OF FIRING CABLE



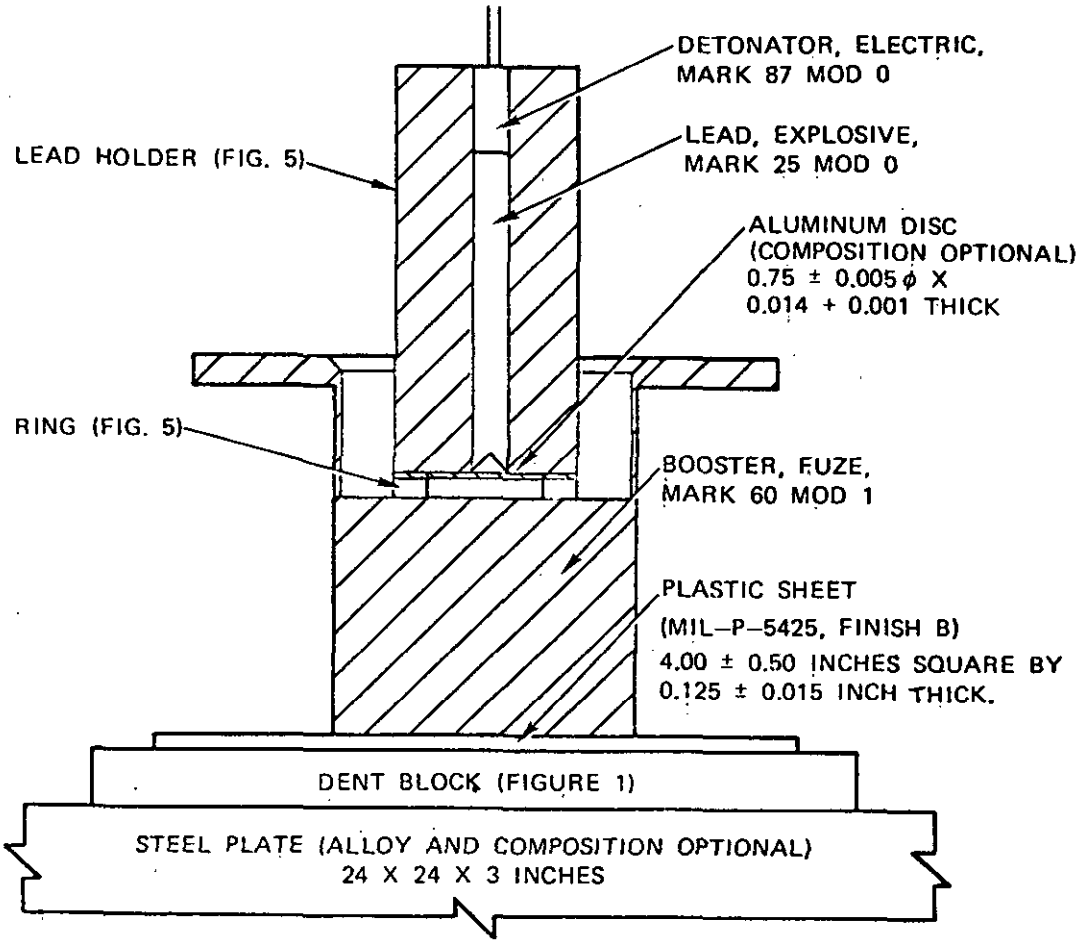
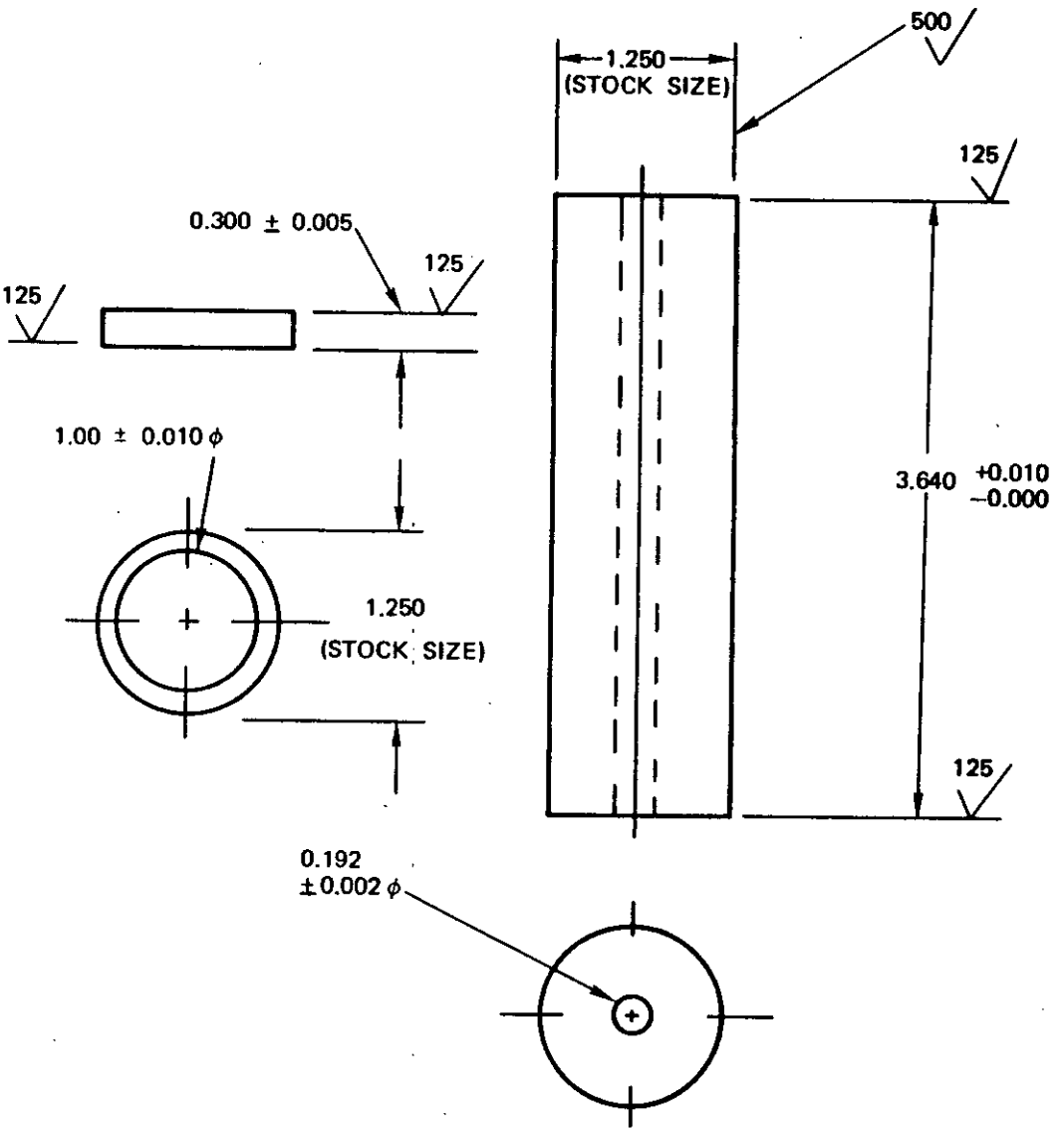


FIGURE 4. Function and output test setup.

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NOTES:

1. MATERIAL SHALL BE STEEL, SAE 1018 THROUGH 1035, COLD– OR HOT–ROLLED.
2. DIMENSIONS ARE IN INCHES.

FIGURE 5. Lead holder and ring test assembly.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE
MIL-B-85349(AS), Booster, Fuze, MARK 60 MOD 1

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 Replaces edition of 1 Jan 72 which may be used. S/N 0102-LF-001-4260

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